Higher National Diploma in Engineering (Electrical and Electronics)

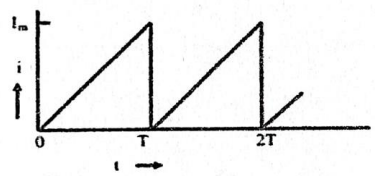
First Year, First Semester Examination — 2022

22EN1212: Electrical and Electronics Engineering Fundamentals - II

Question 1: AC Fundamentals

(i) Write the mathematical expression for the **average** and **root mean square (r.m.s.) calculation** for random a.c signal. (4 marks)

(ii) Obtain **average** and **r.m.s. values** for the **triangular current waveform** are given below. . (6 marks)



Triangular current waveform

(iii) An alternating voltage is given by $V(t) = 325\ \sin(314.28t)$ Volt.

Find: Peak to Peak value Root Mean Square (r.m.s.) voltage Frequency Instantaneous value of voltage when $t = 3ms$

(iv) Two a.c. voltages are given by $V_1(t) = 120\sin(\omega t)$ and $V_2(t) = 200\sin(\omega t - \pi/4)$ volts. Obtain the sinusoidal expression for $V_1(t) + V_2(t)$ by resolution of phasors. Whether $V_2(t)$ is leading or lagging $V_1(t)$? Give the reason for your answer.

Q2.

(i) A $40\ \mu\mathrm{F}$ capacitor is connected in parallel with a $100\ \Omega$ resistor across a $240\ \mathrm{V}, 50\ \mathrm{Hz}$ supply. Calculate the following: (a) The current in each branch (b) The supply current (c) The circuit phase angle between supply voltage and current (d) The circuit impedance (e) The power dissipated in the circuit (f) The apparent power in the circuit (g) The power factor of the supply (14 marks)

(ii) Define the following for resonant circuits. (a) Resonance frequency (fr) (b) Quality factor $(\mathrm{Q}$-factor) (c) Bandwidth (BW) (3 marks)

(iii) A coil having a resistance of $20\ \Omega$ and an inductance of $200\ \mathrm{mH}$ is connected in series with a $60\ \mu\mathrm{F}$ capacitor across a $230\ \mathrm{V}$ supply. (a) At what frequency does resonance occur? (b) Find the current flowing at the resonance frequency. (c) Determine the bandwidth and Q-factor of the circuit. (8 marks)

Q3.

(i) (a) What are the advantages of a three-phase system over a single-phase system? (State two advantages.)

(b) Compare the Star and Delta connections in the 3-phase system. (4 marks)

(ii) Three coils have resistance of $5\ \Omega$, and inductive reactance of $10\ \Omega$ and are connected in star to a $400\ \mathrm{V}$, 3-phase supply. Calculate the following: (a) The line and phase voltages (b) The line and phase currents (c) Apparent power delivered by the supply (d) Power factor of the load (e) How do you measure the active power in a 3-phase system, using the two-wattmeter method? (f) What are the readings of each wattmeter for the above 3-phase circuit? (11 marks)

(iii) (a) State two advantages and one disadvantage of the PMMC (Permanent Magnet Moving Coil) instrument used in the construction of voltmeters and ammeters. (b) Briefly explain the accuracy and precision of the measurement of electrical quantities. (6 marks)

(iv) A moving-coil instrument gives full-scale deflection with $20\ \mathrm{mA}$ and has a resistance of $10\ \Omega$.Calculate the resistance required: (a) in parallel to enable the instrument to read up to $1\ \mathrm{A}$. (b) in series to enable it to read up to $10\ \mathrm{V}$. (4 marks)

Q4.

(i) (a) Briefly explain the necessity of a transformer in the power system. (b) How does a transformer operate? (c) What is the advantage of using an isolation transformer in the power circuits? (d) What are the losses that took place in the transformer while operating on load? Briefly explain them. (e) Define the voltage regulation of the transformer. Why is it important in the transformer? (8 marks)

(ii) (a) What are the common losses in any type of rotating machine? (b) What is the condition to be satisfied by any type of rotating machine to operate at its maximum efficiency? (c) State the difference between $\mathrm{D.C.}$ generator and the $\mathrm{D.C.}$ motor (d) Compare the $\mathrm{A.C.}$ synchronous machine with the Induction machine. (6 marks)

(iii) A $5\ \mathrm{kVA}$ single-phase transformer has a turn ratio of $10:1$ and is fed from $2\ \mathrm{kV}$ supply. Calculate the followings: (a) Approximate values of primary and secondary currents. (b) The $\mathrm{m.m.f}$ (Magnetic motive force) of secondary which has 80 turns of winding. (c) The approximate number of turns in the primary winding (d) The minimum load resistance that can be connected across the secondary winding to get full load kVA. (e) What is the voltage regulation of the transformer if the voltage at load is $195\ \mathrm{V}$? (7 marks)

Q5.

(i) (a) Covert the following binary numbers into their hexadecimal and decimal equivalent: (I) 10101011101 (II) 1001010111 (b) Decode the following BCD numbers: (I) $01000001\ 01110000\ 00010010\ 10010001\ 10000110$ (II) $01100001\ 01010001\ 01010000\ 01010000\ 00000001$ (c) Distinguish between a binary and a BCD code. Why BCD codes are used for decimal numbers in digital systems?

(ii) (a) Using the theorems of Boolean algebra, prove the following identities: (I) $A \cdot \overline{B} + \overline{A} \cdot B = \overline{A \cdot B + \overline{A} \cdot \overline{B}}$ (II) $\overline{X} \cdot (X + Z) + \overline{Y} + Y \cdot Z = \overline{Y} + Z$ (III) $A \cdot B + A \cdot C + \overline{B} \cdot C = A \cdot B + \overline{B} \cdot C$ (6 marks) (b) Using the theorems of Boolean algebra, reduce the following function and draw the logic circuit using AND, OR and NOT gates only:$F_1(A, B, C, D) = \Sigma(0, 1, 2, 6, 7, 14, 15)$ (6 marks)

(iii) (a) What are the differences between conventional and non-conventional sources of energy? (b) Briefly explain the electric shock and the causes of electric shock. (c) Briefly explain of the earthing systems used in electrical wiring. (5 marks)

(iv) (a) What is IEE wiring regulations and the latest version of these regulations? (b) What are the basic steps or requirements in wiring a residential house? (c) What is the difference between RCCB and MCB? (d) Why do we need 2-way switch in-house electrical wiring? Draw circuit diagram for 2-way switch in detail. (8 marks)